

Conjectures Related to *Lights Out*

William Boyles

August 14, 2026

1 Definitions

Let *Lights Out* be a game played on a simple graph $G = (V, E)$ where clicking a vertex v toggles the on/off state of v and its neighbors. One wins the game by finding a sequence of clicks that turns off all the lights.

Definition 1. *The Most Clicks Problem (MCP) asks, for playing on some graph G , what is the most clicks needed to solve any solvable starting configuration, assuming that one solves each configuration in as few clicks as possible.*

That is, what is the most number of moves one needs to be able to solve any solvable initial configuration? This question is akin to looking for “God’s Number” on a Rubik’s Cube.

Definition 2. *Let $d(n) = \dim(\ker(A + I))$ over the field $\mathbb{Z}_2$ where A is the adjacency matrix of G an $n \times n$ grid.*

You may also hear $d(n)$ referred to as the *nullity* of the $n \times n$ grid. The right-hand side of the equation applies for any graph, not just grids. Elements of the null space correspond to sets of clicks that have no net effect on the states of the vertices when applied. These elements also correspond exactly to subsets of V such that all vertices have an even number of elements from the subset in their closed neighborhood (themselves and their neighbors). Thus, these elements of the null space may be referred to as *null patterns*, *quiet patterns*, or *even dominating sets*.

Definition 3. *Let $F_n(x)$ be the degree n polynomial in the ring $\mathbb{Z}_2[x]$ such that*

$$F_n(x) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ xF_{n-1}(x) + F_{n-2}(x) & n > 1 \end{cases}.$$

These polynomials become useful in calculating $d(n)$. It is known that $d(n) = \deg \gcd(F_{n+1}(x), F_{n+1}(x+1))$.

2 Conjectures

We’d like to extend our existing formulas for $d(n)$, which we have attacked so far by looking at the prime factorization of $n+1$ or the factorization of F_n . Although we list many below, we suspect that are many more similar relations with prime factorization to be found.

Conjecture 1. For all $n, m, o \in \mathbb{N}$, the following hold:

$$\begin{aligned}
d(3^n * 5^m - 1) &= 4 \\
d(3^n * 7^m - 1) &= \begin{cases} 0 & n = 1 \text{ or } m = 0 \\ 24 & n \geq 1 \text{ and } m > 0 \end{cases} \\
d(3^n * 11^m - 1) &= 20 \\
d(3^n * 13^m - 1) &= 0 \\
d(3^n * 17^m - 1) &= 8 \\
d(3^n * 19^m - 1) &= \begin{cases} 0 & n \leq 1 \text{ or } m = 0 \\ 36 & n = 2 \text{ and } m > 0 \\ 252 & n \geq 3 \text{ and } m > 0 \end{cases} \\
d(5^n * 7^m - 1) &= 4 \\
d(5^n * 11^m - 1) &= 4 \\
d(5^n * 13^m - 1) &= 28 \\
d(5^n * 17^m - 1) &= 12 \\
d(5^n * 19^m - 1) &= 4 \\
d(7^n * 11^m - 1) &= 0 \\
d(7^n * 13^m - 1) &= 0 \\
d(7^n * 17^m - 1) &= 8 \\
d(7^n * 19^m - 1) &= 0 \\
d(11^n * 13^m - 1) &= 0 \\
d(11^n * 17^m - 1) &= 8 \\
d(11^n * 19^m - 1) &= 0 \\
d(13^n * 17^m - 1) &= 8 \\
d(13^n * 19^m - 1) &= 0 \\
d(17^n * 19^m - 1) &= 8.
\end{aligned}$$

Products containing 3 seem to have a much more erratic behavior than products without. Notice that these formulas mostly don't depend on the exponents of the primes. Recall that we have already proven a recurrence that allows us to add a power of 2 to any of the products.

We also have conjectures about $d(p^k - 1)$ for p a prime.

Conjecture 2. Let p be a Wieferich prime. Then

$$d(p^k - 1) = d(p - 1).$$

We already already proven this same formula, for non-Wieferich primes.

In general, we've noticed that for odd a , $d(a^k - 1) = d(a - 1)$. We already proved this formula true when a is a non-Wieferich prime. Conjecture 2 extends this to all primes. Conjecture 1 extends this to many products of primes. We have however found some exceptional cases. For example: $d(21 - 1) = 0$, $d(21^2 - 1) = 24$ and $d(57 - 1) = 0$, $d(57^2 - 1) = 36$, $d(57^3 - 1) = 252$.

Conjecture 3. For all $n \in \mathbb{N}$, there exists some $k_0 \in \mathbb{N}$ such that for all integers $k \geq k_0$,

$$d(n^k - 1) = d(n^{k_0} - 1).$$

Moreover, when $3 \nmid n$, $k_0 = 1$.

What remains then is to classify all exceptional values and understand what mechanism describes them.

Conjecture 4. $d(p-1) \neq 0$ for prime $p > 7$ if and only if there exists $1 \leq n < p-1$ such that $2^n \equiv \pm 1 \pmod{p}$.

This fact is already known to be true for all Fermat primes and Mersenne primes.

We know that $d(n) = 26$ is the smallest even impossible value for d . We also know that if a is an impossible even value of d that $2a+2$ is also impossible. We would like to better classify and understand impossible values.

Conjecture 5. Let I be the set of even values that are never values of $d(n)$. Let a family of impossible values be from the recurrence $a_{n+1} = 2a_n + 2$. Then the following are true

1. $26 \in I$ (already known)
2. I is infinite (already known)
3. $34 \in I$ (implies $70, 142, 286, \dots \in I$)
4. I has infinitely many “seed” values. That is, those that are not implied to be in I because a smaller value is also in I .

Our method for solving the MCP for nullity 2 grids of size $6k-1$ involved getting an exact solution to a integer linear program by noticing all constraints could be tightly satisfied at once. This allowed us to give a parabola in terms of k that solved the MCP. However, this does not seem to work for higher nullities. We suspect that the answer to the MCP is still a parabola though. Thus, we can calculate the answer to the MCP for 3 boards of the same nullity in the same family (i.e. size is same mod some number) and derive the parabola.

Conjecture 6. Regarding the answer to the MCP for an $n \times n$ grid,

1. If $n = 6k-1$ and $d(n) = 2$, then the answer to the MCP is $26k^2 - 12k + 1$ (already known).
2. All nullity 2 grids are of size $6k-1$ (already known).
3. If $n = 10k-6$ and $d(n) = 4$, then the answer to the MCP is $17k^2 - 10k$.
4. All nullity 4 grids are of size $10k-6$
5. If $n = 24k-13$ and $d(n) = 6$, then the answer to the MCP is $88k^2 - 24k + 1$.
6. All nullity 6 grids are of size $24k-13$.
7. If $n = 40k-31$ and $d(n) = 8$, then the answer to the MCP is $60k^2 - 20k - 3$.
8. If $n = 34k-18$ and $d(n) = 8$, then the answer to the MCP is $161k^2 - 34k - 3$.
9. All nullity 8 grids are either of size $40k-31$ or $34k-18$. (False for $d(69) = 8$)
10. If $n = 60k-31$ and $d(n) = 10$ then the answer to the MCP is $506k^2 - 60k - 3$.
11. All nullity 10 grids are of size $60k-31$.
12. If $n = 340k-256$ and $d(n) = 12$, then the answer to the MCP is $3761k^2 - 169k - 13$.
13. All nullity 12 grids are of size $340k-256$.
14. ...

Notice that there are different “families” associated with each nullity. These families correspond to a certain size of board. Notice that one nullity might have multiple families. It seems that the smallest board in each family appears in OEIS sequence A118142. We suspect that the parabolas giving the answer to the MCP for a given family always have integer coefficients.

We also have a conjecture about the maximum values of $d(n)$.

Conjecture 7.

$$d(n) \leq \frac{4(n+1)}{5},$$

with equality when $n = 5 \cdot 2^k - 1$.

Our results have already known the equality case, but not that some larger proportion can exist. As far as we know, the only other proof on bounds of $d(n)$ known to date is that $d(4) = 4$ is the only case where $d(n) = n$.